

Oil Comment: Estimating Large Demand Destruction

- The larger-than-expected decline in global oil demand implies that the risks to our Brent/WTI 2026Q4 price forecasts of \$90/83 remain two-sided with significant upside price risks from potentially more persistent Mideast supply losses but also meaningful price downside from weaker demand. We estimate 4-5mb/d of global oil demand destruction in April across three different approaches (Exhibit 1), as the Hormuz shock appears to have reduced global demand by 4-5% relative to a no war counterfactual.
- The first approach estimates global apparent refined product demand as global refinery runs minus estimated draws in global refined product inventories. This approach estimates just under 4mb/d of global oil demand destruction (Exhibit 2) after accounting for a 2mb/d hit to oil demand not produced by refineries (i.e. petrochemical feedstocks).
- The second approach leverages available high frequency measures of oil demand and changes in retail fuel prices (Exhibit 7) combined with an estimate of the price sensitivity of demand elsewhere, and estimates just over 5mb/d of global oil demand destruction. The high-frequency demand measures include April retail fuel sales reports, which are very weak in China (Exhibit 3) and soft in Western Europe, and our solid US gasoline demand nowcast (Exhibit 4).¹
- The third approach averages recent estimates from other forecasters and trading houses both estimating global demand destruction at around 4mb/d (Exhibit 5 Exhibit 6).

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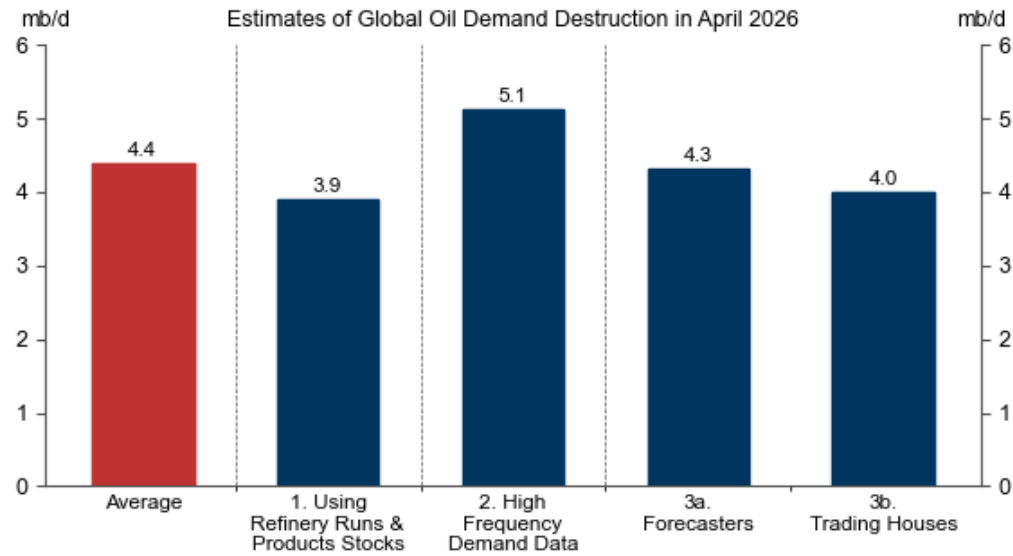
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¹ We use high-frequency demand measures for gasoline in China, the US, Europe, and India and for diesel in Europe. We estimate gasoline demand elsewhere combining changes in retail gasoline prices and an estimated elasticity of -0.25, based on academic estimates and evidence from the Hormuz shock so far. Finally, for other products we use an average of demand estimates across IEA, S&P, and Woodmac.

Three Different Approaches All Generate a 4-5mb/d Estimate of Global Oil Demand Destruction

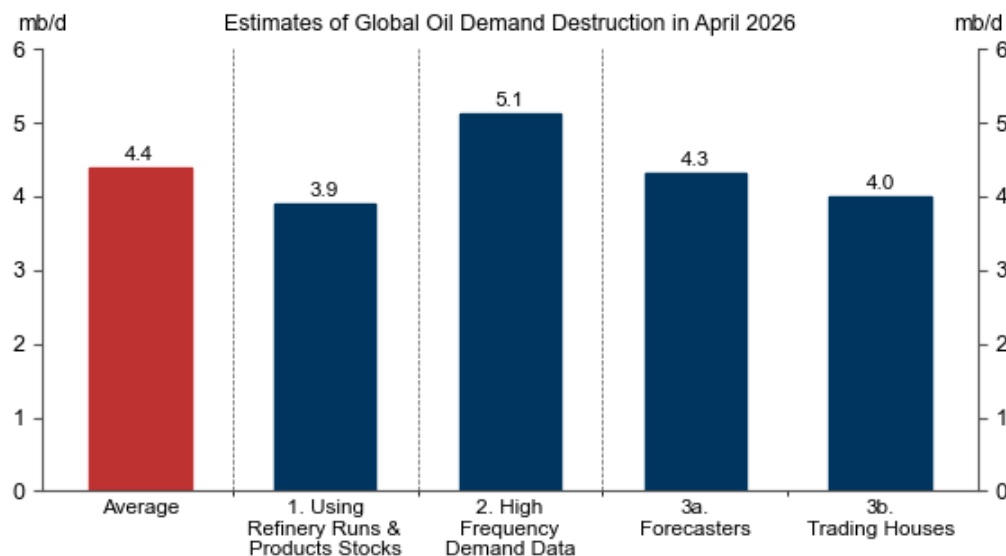


We assume that global oil demand would have grown at an annualized 1mb/d pace without the Hormuz shock.

Source: Goldman Sachs Global Investment Research

Estimating Large Demand Destruction

Exhibit 1: Three Different Approaches All Generate a 4-5mb/d Estimate of Global Oil Demand Destruction

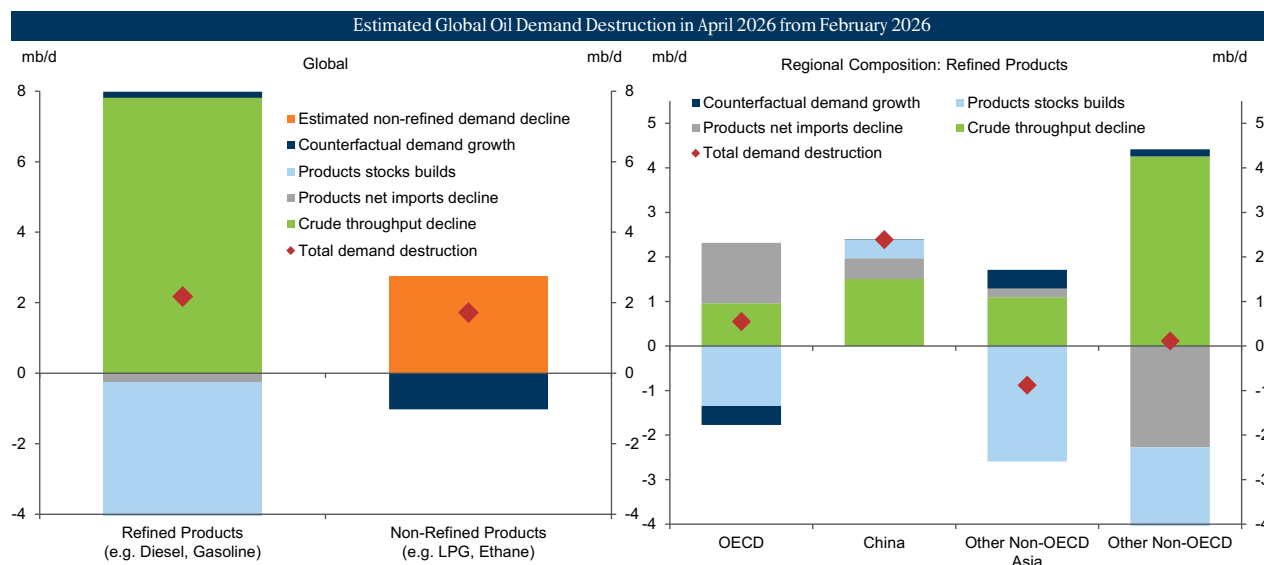


We assume that global oil demand would have grown at an annualized 1mb/d pace without the Hormuz shock.

Source: Goldman Sachs Global Investment Research

Approach 1: Using Refinery Runs and Product Stocks

Exhibit 2: Our Approach Using Refinery Runs and Product Stocks Implies Just Under 4mb/d of Demand Destruction in April, With Roughly Equal Contributions From Refined and Non-Refined Products



We estimate non-refined demand decline as an average of April declines in LPG and ethane demand in the IEA and S&P Global May balances. Estimated product stocks draws are the sum of visible and invisible draws.

Source: IEA, Kpler, S&P Global, Woodmac, Goldman Sachs Global Investment Research

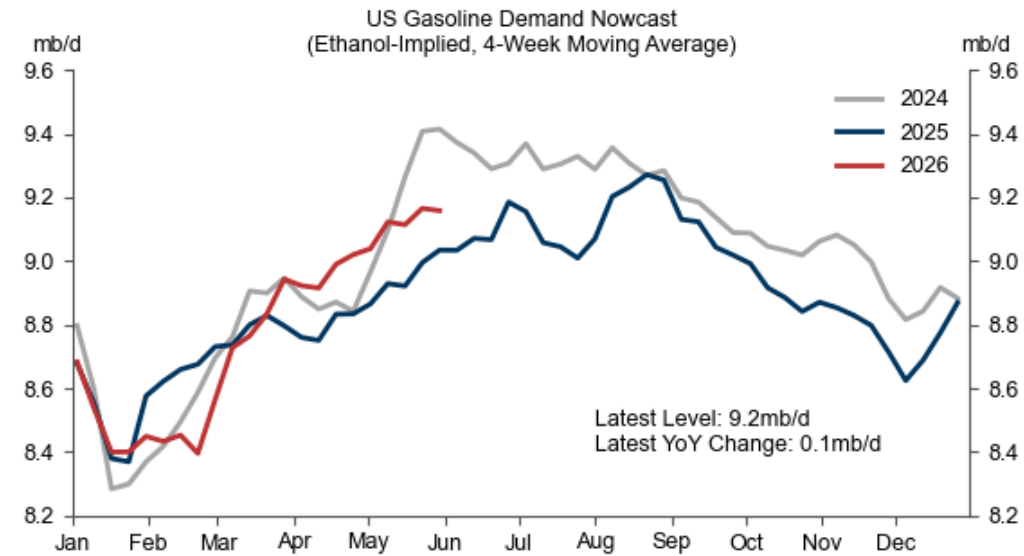
Approach 2: High Frequency Demand Data

Exhibit 3: We Estimate a Just Over 20% Year-Over-Year April Decline in China Gasoline and Related Products Retail Sales Volumes



Source: Wind, NBS, Goldman Sachs Global Investment Research

Exhibit 4: Our US Gasoline Demand Nowcast Remains Solid



Source: EIA, Goldman Sachs Global Investment Research

Approach 3: Wisdom of the Crowds

Approach 3a: Forecasters

Exhibit 5: Averaging Recent Estimates Across Other Forecasters Implies Global Demand Destruction Estimated at 4.3mb/d With Much of the Weakness Concentrated in Asia and the Middle East

External Estimates of April Global Oil Demand (mb/d)								
	Demand Destruction	Year-Over-Year Change						
	Global	Global	US	Europe	China	Asia ex China	Middle East	Rest of World
IEA		-2.5	0.2	-0.6	-0.2	-1.0	-0.7	-0.2
S&P		-4.1	0.5	-0.5	-1.2	-0.9	-1.5	-0.3
Woodmac		-3.4	-0.4	-0.4	-0.4	-1.3	-0.7	-0.1
Average Across Estimates	-4.3	-3.3	0.1	-0.5	-0.6	-1.1	-1.0	-0.2

We estimate demand destruction as the year-over-year change in global oil demand minus the 1mb/d estimated annualized trend in global oil demand growth.

Source: IEA, S&P Global Market Intelligence, Wood Mackenzie, Goldman Sachs Global Investment Research

Approach 3b: Trading Houses

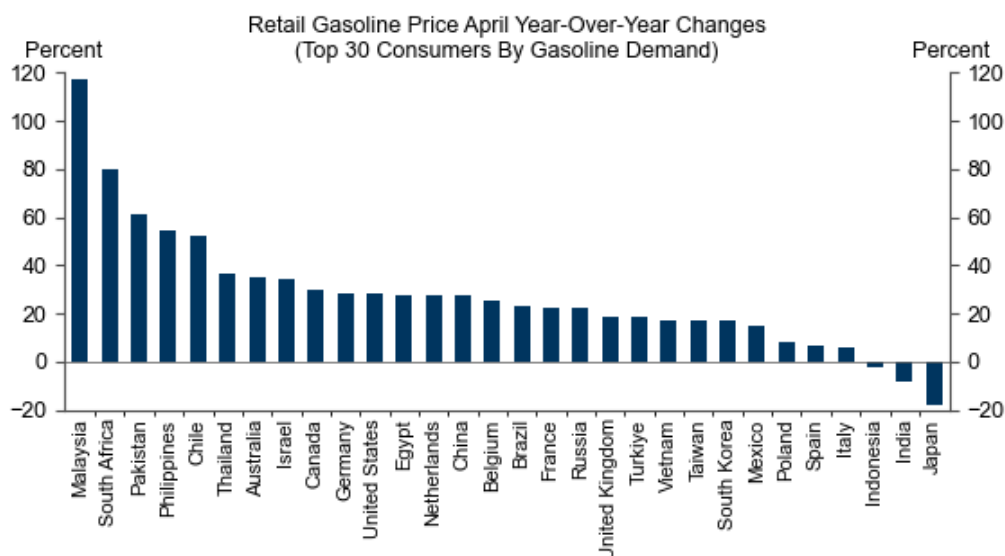
Exhibit 6: Averaging Recent Estimates Across Trading Houses Implies Global Demand Destruction Estimated at 4mb/d

Region/Type of Oil	Trading House	Estimated Demand Destruction (mb/d)	Time Period	Date That Estimate Was Made	Comment
Global Total Oil	Vitol	4	March-April	21-Apr	
	Gunvor	2.5	April	22-Apr	
	Gunvor	5	May	22-Apr	
	Vitol	4-5	"When oil prices were closer to \$110"	2-Jun	Mostly in Asia and Africa.
	Average	4			
China Gasoline	Mercuria	1	2026	21-Apr	

Source: Reuters, Bloomberg, Goldman Sachs Global Investment Research

Appendix Charts

Exhibit 7: Our Second Approaches Leverages Changes in Retail Gasoline Prices in Countries Where No High Frequency Demand Proxies Are Available



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 8: Averaging Recent Estimates Across Other Forecasters Implies Global Demand Destruction Estimated at 4.3mb/d With Much of the Weakness Concentrated in Petrochemical Feedstocks

External Estimates of April Global Oil Demand (mb/d)							
	Demand Destruction	Year-Over-Year Change					
	Global	Global	LPG, Ethane, Naphtha	Motor Gasoline	Jet and Kerosene	Gasoi// Diesel	Other
IEA		-2.5	-2.0	-0.1	-0.3	-0.4	0.3
S&P		-4.1	-1.6	-0.4	-0.3	-0.9	-0.9
Woodmac		-3.4	-1.3	-0.9	-0.4	-0.6	-0.2
Average Across Estimates	-4.3	-3.3	-1.6	-0.5	-0.3	-0.6	-0.2

We estimate demand destruction as the year-over-year change in global oil demand minus the 1mb/d estimated annualized trend in global oil demand growth.

Source: IEA, S&P Global Market Intelligence, Wood Mackenzie, Goldman Sachs Global Investment Research

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